STATIC MAGNETIC FIELDS

- Magnetic force
- Torque
- Ampere's force Law

$$\vec{F}_m = q\vec{u} \times \vec{B} \quad (N)$$

$$d\vec{f}_m = -\hat{N}_e \hat{S} |dl| \vec{u} \times \vec{B}$$

$$d\vec{f}_m = I d\vec{l} \times \vec{B} \quad (N)$$

When we have two circuits

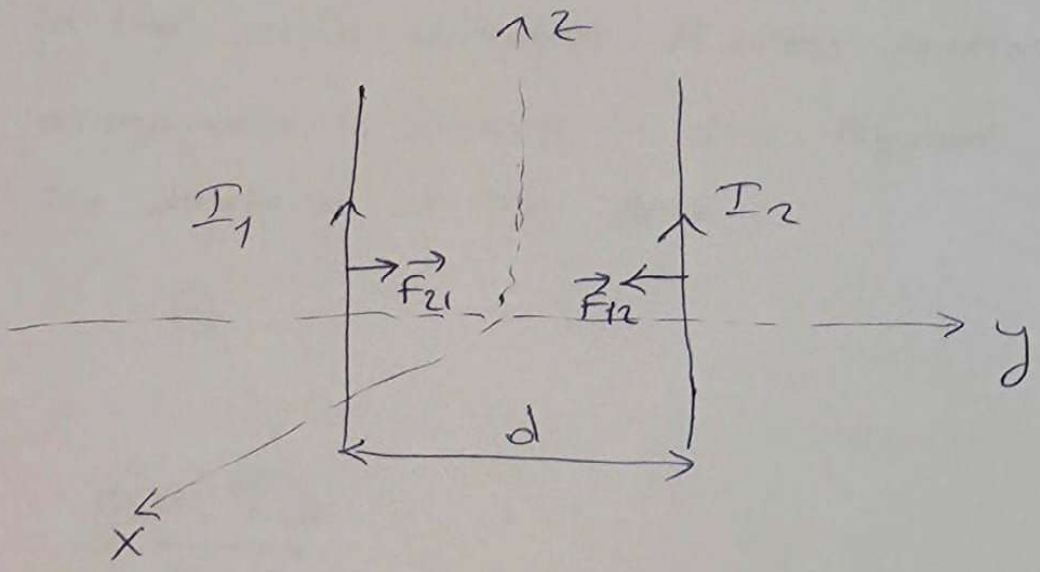
$$\vec{F}_{12} = I_2 \oint_{C_2} d\vec{l}_2 \times \vec{B}_{12}$$

$$\vec{B}_{12} = \frac{\mu_0 I_1}{4\pi} \oint_{C_1} \frac{d\vec{l}_1 \times \vec{a}_{R12}}{R_{12}^2}$$

if the wires are infinitely long and cylindrical symmetry exists $\vec{B}_{12} = \vec{a}_{R12} \frac{\mu_0 I}{2\pi d}$

Ampere's law of force

$$\vec{F}_{12} = \frac{\mu_0}{4\pi} I_2 I_1 \oint_{C_2} \oint_{C_1} \frac{d\vec{l}_2 \times (d\vec{l}_1 \times \vec{a}_{R12})}{R_{12}^2} \quad (N)$$



Force of attraction between wires carrying currents in the same direction

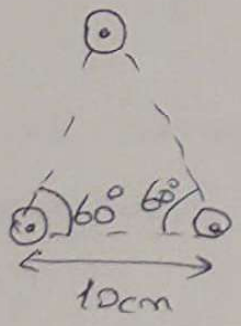
[unlike the force btw two charges of the same polarity → repulsion]

Magnetic dipole moment $\vec{m} = \vec{a}_n I S$

Torque $\vec{T} = \vec{m} \times \vec{B}$

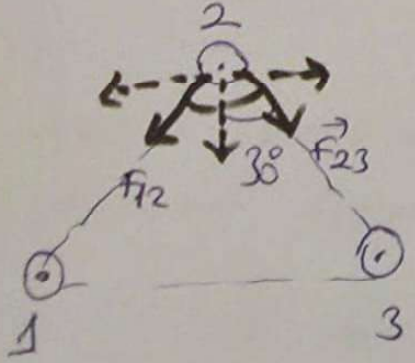
Ex / Pr. 5.18

Calculate the force per unit length on each of three equidistant, infinitely long, parallel wires (10cm) apart, each carrying a current of 25 (A) in the same direction. A cross section of the arrangement is shown in the figure. Specify the direction of the force.



$I_1 = I_2 = I_3 = 25\,A$ $\quad d = 10\,cm$

Force per unit length on wire 2 :



$$\frac{F}{\ell}(2) = 2 \cdot \frac{F_0}{L} \cos 30$$

$$\frac{F_0}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} \Rightarrow \frac{F_0}{L} = \frac{4\pi \cdot 10^{-7} \cdot 25 \cdot 25}{2\pi \cdot 0,1}$$

$$\frac{F_0}{L} = 1,25 \times 10^{-3}\ N/m$$

All 3 wires are attracting!

$$\frac{F(2)}{\ell} = 2 \cdot 1,25 \times 10^{-3} \times (0,866)$$

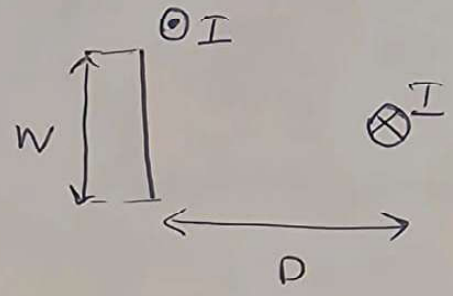
$$= 2,17 \times 10^{-3}\ N/m$$

Forces on all 3 wires are of equal magnitude and toward the center of the triangle.

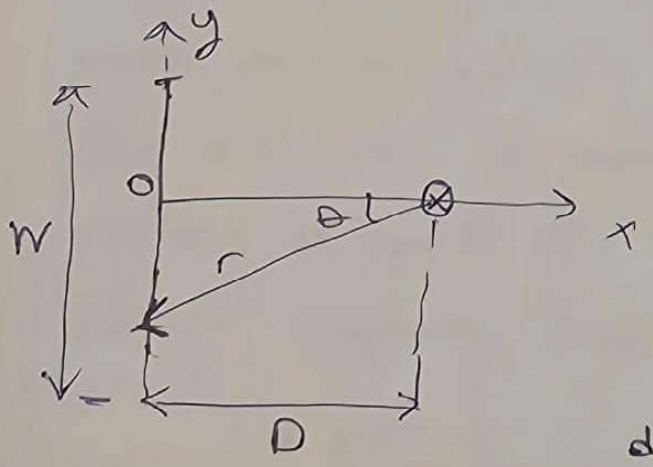
Pr. 5.19

Ex

The cross section of a long thin metal strip and a parallel wire is shown in the figure. Equal and opposite currents I flow in the conductors. Find the force per unit length on the conductors.



Equal currents I flow in opposite directions, so the force is repulsive.



A small element of width dy carries $dI = \frac{I}{W} dy$

The distance $r = \sqrt{D^2 + y^2}$

$$d\left(\frac{f}{L}\right) = \frac{\mu_0 I dI}{2\pi r}$$

radical components cancel by symmetry

horizontal component $d\left(\frac{f_x}{L}\right) = \frac{\mu_0 I dI}{2\pi r} \cos\theta$

$\cos\theta = \frac{D}{r} \Rightarrow \frac{d(f_x)}{L} = \frac{\mu_0 I^2}{2\pi W} \frac{D\, dy}{D^2 + y^2}$

Integrate over the strip

$$\frac{F}{L} = \frac{\mu_0 I^2}{2\pi W} \int_{-W/2}^{W/2} \frac{D\,dy}{D^2+y^2}$$

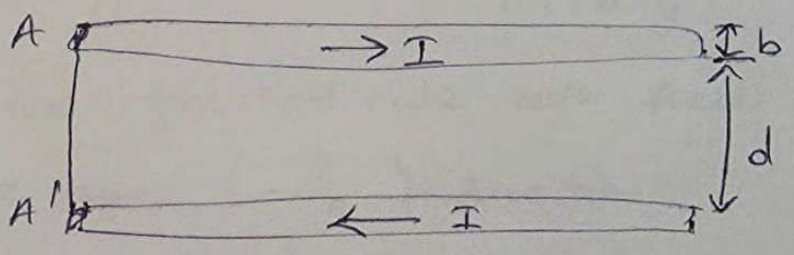
$$\int \frac{D\,dy}{D^2+y^2} = \tan^{-1}\left(\frac{y}{D}\right)$$

$$\frac{F}{L} = \frac{\mu_0 I^2}{\pi W} \tan^{-1}\left(\frac{W}{2D}\right) \quad (N/m)$$

Repulsive force

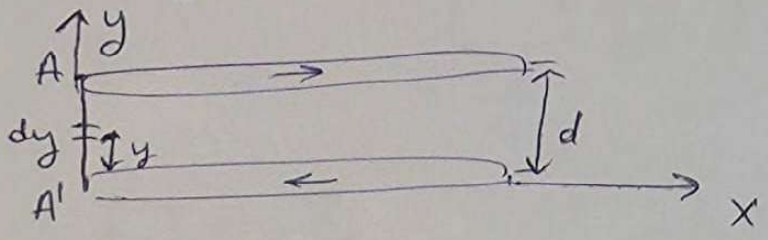
Pr. 5.20
Ex

The bar AA' in the figure serves as a conducting path for the current I in two very long parallel lines. The lines have a radius b and are spaced at a distance d apart. Find the direction and the magnitude of the magnetic force on the bar.



- Two very long parallel cylindrical conductors
- each conductor has radius b
- the center to center spacing btw them is d,
- AA' connects the two conductors

The bar extends from (y-axis upward along the bar)
y=b to y=d-b



for a very long straight conductor

$$\vec{B} = \frac{\mu_0 I}{2\pi r}$$

So the two contributions are

$$B_1 = \frac{\mu_0 I}{2\pi y} \quad , \quad B_2 = \frac{\mu_0 I}{2\pi (d-y)}$$

Using the right-hand rule, both fields are directed into the page $(-\vec{a}_z)$ direction.

$$\vec{B}(y) = -\vec{a}_z \frac{\mu_0 I}{2\pi} \left(\frac{1}{y} + \frac{1}{d-y} \right)$$

A differential element of the bar is

$$d\vec{\ell} = \vec{a}_y \, dy$$

$$d\vec{F} = I \, d\vec{\ell} \times \vec{B}$$

$$d\vec{F} = I(\vec{a}_y \, dy) \times \left[-\vec{a}_z \frac{\mu_0 I}{2\pi} \left(\frac{1}{y} + \frac{1}{d-y} \right) \right]$$

Since

$$\vec{a}_y \times (-\vec{a}_z) = -\vec{a}_x$$

$$d\vec{F} = -\vec{a}_x \frac{\mu_0 I^2}{2\pi} \left(\frac{1}{y} + \frac{1}{d-y} \right) dy$$

$$\vec{F} = -\vec{a}_x \frac{\mu_0 I^2}{2\pi} \int_b^{d-b} \left(\frac{1}{y} + \frac{1}{d-y} \right) dy$$

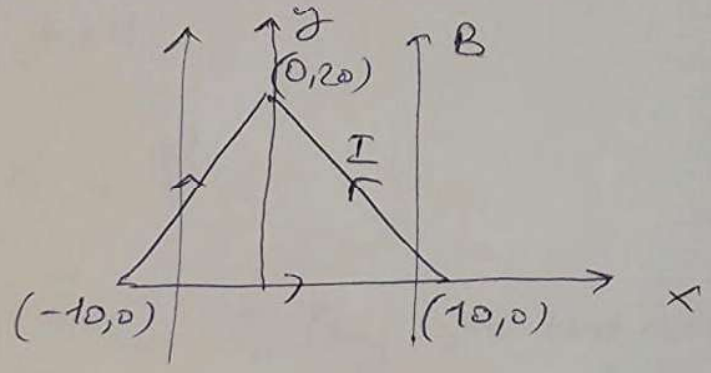
$$\int \frac{dy}{y} = \ln y \qquad \int \frac{dy}{d-y} = -\ln(d-y)$$

$$\vec{F} = -\vec{a}_x \frac{\mu_0 I^2}{\pi} \ln\left(\frac{d}{b} - 1 \right)$$

Pr. 5.21

Ex

A d.c. current $I = 10A$ flows in a triangular loop in the xy plane as in the figure. Assuming a uniform magnetic flux density $\vec{B} = \vec{a}_y 6(mT)$ in the region, find the forces and torque on the loop. The dimensions are in cm.



Force on each side

a) Bottom side $(-10,0) \rightarrow (10,0)$ cm

$\vec{L}_1 = 20cm\,\vec{a}_x = 0,2\vec{a}_x\ m$

$\vec{F}_1 = I\vec{L}_1 \times \vec{B} = 10.(0,20\vec{a}_x) \times (6\times10^{-3}\vec{a}_y)$

$\vec{F}_1 = 0,012\,\vec{a}_z\ N$

b) Right side $(10,0) \rightarrow (0,20\,cm)$

$\vec{L}_2 = (-10\vec{a}_x + 20\vec{a}_y)\,cm = (-0,10\vec{a}_x + 0,20\vec{a}_y)\,m$

$\vec{F}_2 = I\vec{L}_2 \times \vec{B} = 10(-0,10\vec{a}_x + 0,20\vec{a}_y) \times (6\times10^{-3}\vec{a}_y)$

$\vec{F}_2 = -0,006\,\vec{a}_z\ N$

c) Left side $(0,20) \rightarrow (-10,0)$ cm

$\vec{L}_3 = (-0,10\vec{a}_x - 0,20\vec{a}_y)\,m \qquad \vec{F}_3 = I\vec{L}_3 \times \vec{B}$

$\vec{F}_3 = 10(-0,10\vec{a}_x - 0,20\vec{a}_y) \times (6\times10^{-3}\vec{a}_y)$

$$\vec{F}_3 = -0,0006\,\vec{o}_2\ N$$

$$\vec{F}_{net} = \vec{f}_1 + \vec{f}_2 + \vec{f}_3$$

$$= 0,012\,\vec{o}_2 - 0,006\,\vec{o}_2 - 0,006\,\vec{o}_2$$

$$\vec{F}_{net} = 0$$

This is expected for a closed loop in a uniform magnetic field.

$$\vec{C} = \vec{m} \times \vec{B} = \vec{T}$$

$\vec{m} = I A \vec{o}_n \rightarrow \vec{o}_2$ (by right-hand rule, I → counter clockwise)

$\hookrightarrow \left(\frac{1}{2}\,0,2\,0,2\right) = 0,020\,m^2$

$$\vec{T} = (0,20\,\vec{o}_2) \times (6\times10^{-3}\,\vec{o}_y)$$

$$\vec{C} = \vec{T} = -\vec{o}_x\,(12\times10^{-4})\ (N.m)$$